CO1 INTEGRATED ENGINEERING ANALYSIS AND PLANNING ASSIGNMENT

Evolution, cellular planning, spectrum use and interference management

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Course Name :-	5G/6G Communication

Engineering Design Report for the Regional Operator Planning Team

Course outcome assessed: CO1 | Maximum marks: 100

1. Requirements, Scope and Assumptions

A regional operator is planning the next stage of its mobile network. This report analyses the supplied planning data and follows the sequence requirements → analysis → calculations → comparison → recommendation. Each calculation states its equation, units and engineering meaning.

Table 1: Planning data and the design questions they answer

Topic	Supplied data	Design question
Cell geometry	Hexagonal cells, r = 2 km; 19-cell planning cluster; reuse N = 7	Cell and cluster area, reuse distance, coverage versus capacity
Traffic	1,000 subscribers × 0.08 E; C = 100 channels; blocking ≤ 2%	Offered load, Erlang B blocking, channel groups needed
Mobility	60 km/h; 6 s overlap dwell; 1 km cell length; 0.5 s interruption; 1% target-cell unavailability	Handoff rate and call-drop risk
Interference	S = −70 dBm; I_1 = −85 dBm; I_2 = −90 dBm; target SINR 15 dB	C/I, total interference, SINR, mitigation
Spectrum	900 MHz, 3.5 GHz, 28 GHz at 100 m	Wavelength, FSPL, deployment implications

Assumptions used throughout

- **A1 (geometry):** ideal hexagonal tiling on flat, uniform terrain, with base stations at cell centres and equal transmit power.
- **A2 (traffic):** Poisson call arrivals, exponentially distributed holding times, blocked calls cleared, and a large (effectively infinite) user population, as required by the Erlang B model. All 1,000 subscribers are assumed to offer traffic to one pool of channels.
- **A3 (1G channel count):** guard bands, forward/reverse duplexing and control channels are **ignored** in the idealized count.
- **A4 (mobility):** the 1 km value is the route length travelled through a cell between successive handoffs, at a constant 60 km/h.
- **A5 (interference):** no noise power is supplied, so SINR is computed from signal and the two interference terms only. Real SINR can only be equal or lower once thermal noise is added.
- **A6 (propagation):** free-space, line-of-sight path loss with isotropic antennas; no shadowing, building penetration or antenna gain.
- **A7 (handoff failure):** each handoff fails independently with probability 0.01, the target-cell unavailability.
- **A8 (3G access):** the planning data gives only the 5 MHz carrier for UMTS; the WCDMA radio access is added from standard knowledge.

2. Stage 1: Evolution of Mobile Systems (1G to 6G Vision)

Table 2: Generational comparison

Generation	Main service capability	Access / radio technology	Limitation or change that motivated the next generation
1G AMPS	Analog voice only	Analog FDMA; 800 MHz; 30 kHz channels; 10 MHz allocation	Low capacity (one user per channel), poor security, no data service, inefficient spectrum use → move to digital

Generation	Main service capability	Access / radio technology	Limitation or change that motivated the next generation
2G GSM	Digital voice, SMS, encryption	Digital TDMA/FDMA; 900 MHz; 200 kHz carrier; 8 slots	Voice-centric, low circuit-switched data rates → packet data and wideband carriers
3G UMTS	Mobile data, video calling	5 MHz carrier (WCDMA, A8)	Self-interference limits capacity; data rates still modest → all-IP, OFDM-based broadband
4G LTE	All-IP mobile broadband	OFDMA with MIMO	Not built for ultra-low latency or massive device counts; sub-6 GHz spectrum scarce → new use cases and wider bands
5G NR	eMBB, URLLC, massive IoT	Mid-band and mmWave; beamforming	mmWave coverage and blockage; dense, costly deployment; no native sensing or AI → 6G vision
6G (vision)	Holographic communication, tactile internet, ISAC	THz bands; AI-native networking	Extreme path loss, hardware and energy cost at THz; standards not yet defined

Calculations

$$\lambda = c / f = (3 \times 10^8\ m/s) / (800 \times 10^6\ Hz) = 0.375\ m$$

$$N_1G = 10\ MHz / 30\ kHz = 10{,}000\ kHz / 30\ kHz = 333.3 \rightarrow 333\ channels\ (idealized)$$

Practical qualification. The 333 figure ignores guard bands, duplexing and control channels (A3). Cellular service is duplex, so each call needs a forward and a reverse channel. If the 10 MHz were split evenly into 5 MHz in each direction, only about 166 duplex channel pairs would remain. With N = 7 reuse, each cell could use only about 333/7 ≈ 47 of the idealized channels (about 23 duplex pairs), which shows how limited 1G capacity was.

1G to 2G. GSM replaced analog FM voice with digital transmission and TDMA. Eight time slots on a 200 kHz carrier give 200/8 = 25 kHz of spectrum per slot, slightly less than the 30 kHz AMPS channel. In the same 10 MHz, the idealized count would be 50 carriers × 8 = 400 slots, against 333 analog channels. Digital speech coding, error correction and encryption also tolerate a lower carrier-to-interference ratio than analog FM (textbooks quote roughly 18 dB for AMPS and about 9–12 dB for GSM [1]), so tighter frequency reuse is possible. Together these give more capacity, better voice quality and better security.

Synthesis

Across the generations, spectrum moved from a resource divided into narrow channels (1G, 2G) to one used through wide carriers and shared dynamically (3G, 4G). With 5G and the 6G vision the spectrum itself spans sub-GHz to THz bands, and the goal shifts to spectral efficiency, bandwidth, and spatial reuse through MIMO and beamforming. Interference management evolved in step: from fixed frequency planning (N = 7 in 1G and early 2G), to power control and soft handoff in CDMA, to scheduling and inter-cell coordination in LTE, and finally to beam-level and AI-assisted coordination in 5G and 6G. Capacity therefore grew from adding channels, to adding bandwidth, to adding antennas and cells. This is the reason the later stages of this report link reuse, traffic, interference and band choice together.

3. Stage 2: Cell Coverage and Frequency Reuse

Table 3: Hexagonal geometry and reuse results

Quantity	Equation and working	Result
Area of one cell	$A_{cell} = (3\sqrt{3}/2)\, r^2 = 2.598 \times (2\text{ km})^2$	10.39 km²
19-cell cluster area	$A_{19} = 19 \times 10.392\text{ km}^2$	197.45 km²
Reuse fraction	$1/N = 1/7$	0.143 (14.3%)
Reuse distance	$D = r\sqrt{(3N)} = 2\text{ km} \times \sqrt{21} = 2 \times 4.583$	9.165 km
Co-channel ratio	$D/r = \sqrt{(3N)}$	4.58

Each cell may use only one seventh of the total channels. A 19-cell area therefore contains 19/7 ≈ 2.7 reuse clusters, so each frequency group is reused about 2.7 times in the planning area. Co-channel cells are 9.165 km apart, which is 4.58 cell radii.

Hexagonal assumptions and the limit of geometry. The hexagon is the regular shape closest to a circle that tiles a plane with no gaps or overlap. It assumes flat uniform terrain, a base station at each cell centre, equal power, omnidirectional antennas and a uniform path-loss law (A1). **Geometric coverage** is only the area a cell is assigned on the map. **Guaranteed radio coverage** needs a link budget that holds the required signal level at the edge with a chosen probability, after shadowing, fading, terrain, buildings and indoor loss. Real footprints are irregular, so the hexagon is a planning abstraction and not a coverage guarantee.

Table 4: Capacity-expansion techniques

Technique	Effect on coverage and capacity	Interference and handoff effects
Cell splitting	Smaller radius, more cells per area; more channel reuse per km²	Needs lower power and re-planning; handoff rate rises roughly in proportion to 1/radius
Sectoring (e.g. 120°)	Same site; directional antennas; capacity gain by allowing smaller N	Fewer co-channel interferers seen per sector; more sector handoffs; some trunking loss
Microcells	Street or hotspot coverage of a few hundred metres	Strong local reuse; frequent handoffs for fast users; needs macro overlay
Picocells	Indoor or venue coverage, tens of metres	Isolated by walls; little outdoor interference; backhaul required
Femtocells	Home or office; user-deployed	Uncoordinated deployment can create interference; low-power, low-cost

Recommendation. The traffic calculation in Stage 3 shows that capacity is not the immediate limit, whereas the interference result in Stage 5 is 1.2 dB short of target. The most suitable first expansion measure is therefore **120° three-sector antennas at the macro sites**, which tackles both the interference shortfall and future growth without building new sites. The effect on interference is favourable (fewer co-channel interferers), while the effect on handoff is a rise in intra-site sector handoffs. Cell splitting should be held back for later growth, because halving the radius would roughly double the handoff rate from about 1 to about 2 per minute for the same route and speed. Small cells then serve hotspots only.

4. Stage 3: Traffic Capacity Estimate

$$A = U \cdot a = 1{,}000 \times 0.08\,E = 80\,E$$

Each subscriber offers 0.08 E, which is 0.08 × 60 = 4.8 minutes of channel use in the busy hour. The Erlang B blocking probability for A offered erlangs and C channels is evaluated with the stable recursion below, using a short script (Python), with B(A, 0) = 1:

$$B(A, C) = A \cdot B(A, C-1) / [\, C + A \cdot B(A, C-1) \,]$$

Table 5: Erlang B results for A = 80 E

Channels C	Blocking probability	Meets 2% target?
90	2.62%	No
91	2.25%	No
92	**1.92%**	**Yes (smallest C)**
100 (initial base station)	**0.40%**	**Yes**

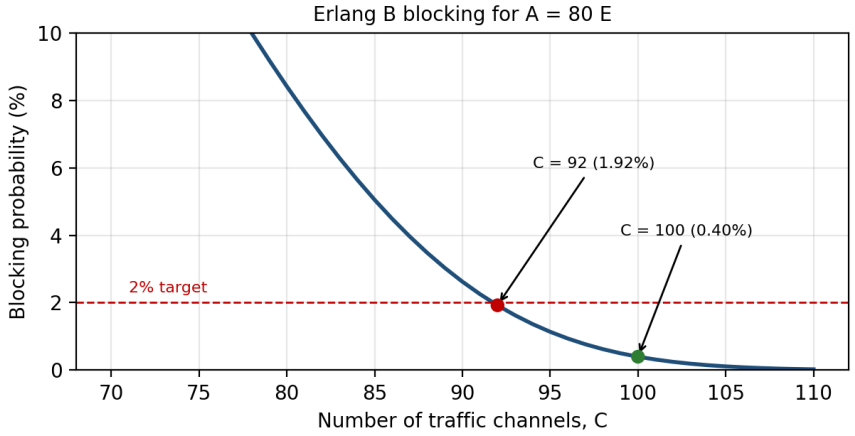


Figure 1: Erlang B blocking probability for A = 80 E. The 2% target is first met at C = 92.

Results. With 100 channels, blocking is **0.40%**, well inside the 2% target. The smallest channel count that meets the target is **C = 92** (91 channels give 2.25%). The number of 100-channel groups needed is ⌈92/100⌉ = **1**, so the initial base station already satisfies the demand. At 100 channels the carried traffic is 80 × (1 − 0.0040) ≈ 79.7 E, roughly 80% channel utilisation. Solving B(A, 100) = 2% gives a maximum offered load of about 88 E, equivalent to about 1,100 subscribers at 0.08 E, so there is around 10% growth headroom.

Trade-offs. The 8 channels between 92 and 100 cost little and provide headroom, and a larger group is more efficient (trunking efficiency) than splitting traffic over smaller groups. Fewer channels would save infrastructure but risk exceeding the blocking target as demand grows. More channels or groups cost spectrum, radios and backhaul for very little blocking improvement, since blocking is already 0.40%.

Why this is only a first-order estimate. It treats the 1,000 users as one pool at one site, which gives no information on where the users are or how the 197 km² area is covered. It also ignores control and signalling channels, sectoring, which would divide the channels into smaller and less efficient pools (for three sectors at 26.7 E each, about 36 channels per sector, 108 in total, are needed for 2%), spatial and time variation of traffic, handoff traffic between cells, and the effect of reuse N = 7 on how many channels each cell can actually be assigned. If the same 1,000 users were spread evenly over the 19

cells, each cell would see only about 4.2 E and need about 9 channels, which shows that the result depends strongly on the spatial distribution assumed (A2). A complete site plan needs coverage, interference and spectrum-availability analysis in addition to this traffic estimate.

5. Stage 4: Mobility and Handoff

$$v = 60\ km/h = 60{,}000\ m / 3{,}600\ s = 16.67\ m/s$$

$$d_overlap = v \cdot t = 16.67\ m/s \times 6\ s = 100\ m$$

The mobile travels 100 m while in the overlap zone, which is 10% of the 1 km cell length (and 10% of the time spent per cell).

$$t_cell = L / v = 1{,}000\ m / 16.67\ m/s = 60\ s \quad \rightarrow \quad handoff\ rate = 60\ s / 60\ s = 1\ handoff\ per\ minute$$

Interpretation of the 1 km length (A4). The data gives r = 2 km, whose diameter would be 4 km, so the 1 km value cannot be a full diameter. I interpret it as the distance along the route between two consecutive handoff points, for example a chord through a cell or a smaller urban cell. Under this reading the user makes **1 hard handoff per minute**. If the true route length were 4 km, the rate would drop to 0.25 per minute, so the cell length is a parameter to verify.

Table 6: Hard, soft and softer handoff

Type	Connection continuity	Resource use	Typical systems
Hard	Break-before-make; brief interruption (0.5 s here)	One link at a time; lowest resource use	GSM, LTE
Soft	Make-before-break; connected to two or more cells at once; very low drop risk	Extra channels in several cells and extra backhaul during overlap	CDMA / UMTS
Softer	Make-before-break between sectors of the same base station	Extra channel elements at one site; no extra backhaul	CDMA / UMTS sectored sites

Call-drop risk estimate

$$P_drop,1 \approx P(target\ cell\ unavailable) = 0.01 = 1\%$$

Under A7, a hard handoff fails if the target cell cannot accept the call at that moment, so the drop risk is about **1% per handoff**. The 0.5 s interruption occupies 0.5/60 = 0.83% of the time between handoffs, which affects service quality but is not itself a drop. For a 3-minute call with 1 handoff per minute, the risk over three handoffs is $1 - 0.99^3 \approx$ **2.97%**.

Limits. The estimate assumes independent handoffs and a constant 1% unavailability. In reality, unavailability is correlated with load, so busy cells will be worse. The 6 s overlap allows several retry attempts, which would reduce the risk, but retries are correlated and cannot be treated as independent. The model also ignores radio failures from fading or low SINR in the target cell, ping-pong handoffs, speed variation, and the interruption time during which the call may fail even if the target cell is available. The 1% per handoff is therefore a simple, approximate figure and not a prediction.

6. Stage 5: Interference and Spectrum-Band Comparison

6.1 Interference calculation

$$C/I = S - I_1 = -70\ dBm - (-85\ dBm) = 15\ dB$$

With the co-channel interferer alone, the ratio is exactly 15 dB, which meets the target with **zero margin**. Adding adjacent-channel interference requires summation in linear power (mW), not in dB:

$$I_1 = 10\text{^}(-85/10) = 3.162 \times 10^{-9}\,mW \qquad I_2 = 10\text{^}(-90/10) = 1.000 \times 10^{-9}\,mW$$

$$I_total = 4.162 \times 10^{-9}\,mW \rightarrow 10\log_{10}(4.162 \times 10^{-9}) = -83.81\,dBm$$

$$SINR = S - I_total = -70 - (-83.81) = 13.81\,dB$$

The result is **13.81 dB**, which is **1.19 dB below the 15 dB target**, so the requirement is **not met**. The adjacent-channel term adds 1.19 dB of interference. Because no noise power is supplied (A5), this is the best case, and real SINR with thermal noise would be lower still. Removing only the adjacent-channel interference would bring the value back to exactly 15 dB, with no margin, so the co-channel term must also be reduced.

Recommended mitigation: 120° sectoring (with downtilt) at the macro sites. It reduces the number of co-channel interferers seen by each user, and in the classical hexagonal analysis with N = 7 it improves S/I by roughly 5 dB (about 18.7 dB omnidirectional against about 24.2 dB sectored, for a path-loss exponent of 4 [1]), which is more than the 1.19 dB required. **Trade-offs:** more antennas and feeder cost, more sector handoffs, and trunking loss because channels are split into smaller pools (Stage 3: 108 channels instead of 92 for the same blocking). Other options, such as larger N, lower power, or better adjacent-channel filtering, either cut capacity per cell or are insufficient on their own.

6.2 Spectrum-band comparison at d = 100 m

$$\lambda = c/f \qquad FSPL\ (dB) = 20\log_{10}(4\pi d/\lambda)$$

Table 7: Wavelength and free-space path loss

Band	Wavelength λ (m)	FSPL at 100 m (dB)	Difference vs 900 MHz (dB)	Half-wave element λ/2
900 MHz	0.3333	71.53	0 (reference)	16.7 cm
3.5 GHz	0.0857	83.32	+11.80	4.3 cm
28 GHz	0.0107	101.38	+29.86	0.54 cm

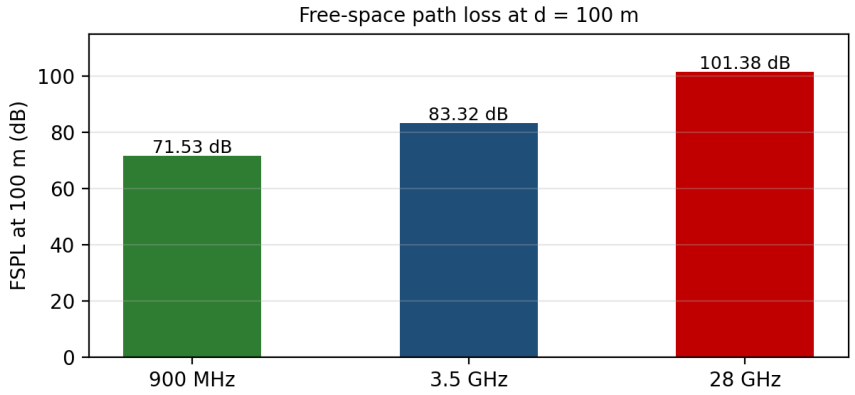


Figure 2: Free-space path loss at 100 m. The difference follows 20 $\log_{10}(f_2/f_1)$: 11.8 dB for 3.5 GHz and 29.9 dB for 28 GHz relative to 900 MHz.

Relative to 900 MHz, the received power at the same distance is lower by a factor of about 15 at 3.5 GHz and about 970 at 28 GHz (isotropic antennas, A6).

Table 8: Deployment implications

Band	Penetration	Antenna size	Coverage	Capacity and typical role
900 MHz	Best; good indoor and rural reach	Largest elements	Widest cells; fewest sites	Limited bandwidth; coverage and mobility layer
3.5 GHz	Moderate; weaker through walls	Compact; practical massive MIMO	Medium cells; often co-sited with low band	Good balance; main 5G eMBB capacity layer
28 GHz	Poor; blocked by walls, foliage, bodies	Very small; large arrays fit in a small area	Short range; line-of-sight dominated	Very wide bandwidth; hotspot, FWA and small-cell capacity

Why path loss alone does not decide performance. Throughput depends on bandwidth and SINR through the Shannon relation $C = B \log_2(1 + SINR)$, not on path loss alone. The extra loss at high frequency comes mainly from the smaller antenna aperture. As elements shrink, many more can be packed into the same physical area, so beamforming gain can recover much of the loss. Higher bands also offer much more bandwidth, and their short range and strong blockage reduce interference and allow dense reuse. As an illustration only (not from the planning data), 100 MHz at 20 dB SINR gives about $100 \times \log_2(101) \approx 666$ Mbit/s, whereas 400 MHz at only 10 dB SINR gives about $400 \times \log_2(11) \approx 1{,}384$ Mbit/s. Real performance also depends on blockage, the link budget, interference, the hardware available and the density of sites, which free-space loss at 100 m does not capture.

7. Stage 6: Spectrum Coordination and Deployment Recommendation

7.1 International spectrum coordination

Table 9: Roles in international spectrum coordination

Body	Distinct role
ITU-R (ITU Radiocommunication Sector)	Global sector of the ITU that manages radio-frequency spectrum and satellite orbits. It runs study groups that develop technical Recommendations and reports (for example the IMT-2020 and IMT-2030 vision documents [2], [3]) and supports the rules in the Radio Regulations.
WRC (World Radiocommunication Conference)	The decision-making conference, held about every three to four years, where member states review and revise the Radio Regulations. It sets frequency allocations between radio services and regions and decides which bands are identified for IMT.
BR (Radiocommunication Bureau)	The administrative and executive arm of the ITU-R, led by an elected Director. It applies the Radio Regulations, processes frequency and satellite network notifications, records assignments in the Master International Frequency Register, assists with coordination and interference cases, and supports the study groups and conferences.

Effect on national regulators and operators. International allocations guide national band plans. When a band is identified for IMT, a national regulator can licence it for mobile use with confidence that equipment, roaming and economies of scale exist, as with the 3.5 GHz range. The regulator must still coordinate use near borders so that its operators do not cause harmful interference to neighbouring countries. For the regional operator, this means that band choice, licence conditions and the power or antenna limits near borders depend on decisions made internationally.

7.2 Recommended staged strategy

Table 10: Phased network plan

Phase	Layer and band	Action and justification from the analysis	Trigger to proceed
1	900 MHz macro layer (LTE-class, 5G-ready), 3-sector sites	Wide coverage and reliable mobility (1 handoff/min, 100 m overlap). One 100-channel group meets the 2% target (92 needed, 0.40% blocking). Sectoring lifts SINR above the 15 dB target.	Drive tests confirm SINR ≥ 15 dB and edge coverage
2	3.5 GHz 5G NR (eMBB) co-sited with Phase 1 sites	Adds bandwidth and capacity. The extra 11.8 dB path loss is offset by massive MIMO beamforming and co-siting. 900 MHz remains the coverage and handoff anchor.	Busy-hour load approaches about 88 E (≈ 1,100 subscribers) or blocking nears 2%
3	28 GHz small cells (micro/pico) at hotspots only	Capacity and URLLC/FWA where demand is dense and line-of-sight exists (29.9 dB more loss than 900 MHz). Fast users stay on the macro layer to limit handoffs.	Demonstrated hotspot demand and fibre backhaul
4	6G readiness	Keep sites, fibre backhaul and software upgradeable for AI-native management and ISAC; follow ITU-R IMT-2030 and WRC outcomes before committing to THz.	Standards and spectrum decisions mature

How the evidence supports the plan. Evolution shows that capacity growth now comes from bandwidth and spatial reuse, not from more narrow channels, so the plan layers low, mid and high bands. The cell and traffic results show that one 100-channel group covers the 80 E demand with margin, so capacity is not the first constraint and expansion can be staged. The mobility result (1 handoff per minute, 1% handoff failure, about 3% risk over a 3-minute call) argues for keeping mobile users on larger cells and using small cells selectively. The interference result (13.81 dB against 15 dB) is the immediate technical problem and is solved first by sectoring. The band comparison shows the coverage versus capacity trade-off that determines the order of the phases, and the ITU process determines which bands are available and under what conditions.

Key assumptions and uncertainties to verify before deployment

- Actual traffic per subscriber (0.08 E), growth rate and the spatial distribution of the 1,000 users, since Erlang B is sensitive to the pooling assumption.
- The true route length per cell (1 km versus r = 2 km), actual handoff failure rates, and whether 0.5 s interruption is acceptable for the services offered.
- Measured noise floor and real interference levels, because SINR here ignored noise and the I_1/I_2 values may vary with time and location.
- Real propagation (shadowing, building loss, foliage, terrain) in place of free-space loss, particularly for 3.5 GHz and 28 GHz.
- Spectrum licence availability, licence conditions and cross-border coordination for each band.
- Device ecosystem support for each band, backhaul availability, and capital and operating costs.

References

[1] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall, 2002 (cellular concepts, frequency reuse, sectoring, trunking and Erlang B, handoff).

[2] ITU-R Recommendation M.2083, “IMT Vision: Framework and overall objectives of the future development of IMT for 2020 and beyond,” 2015.

[3] ITU-R Recommendation M.2160, “Framework and overall objectives of the future development of IMT for 2030 and beyond,” 2023.

[4] International Telecommunication Union, Radio Regulations (edition in force) and ITU Constitution and Convention.

[5] C. E. Shannon, “A mathematical theory of communication,” Bell System Technical Journal, vol. 27, 1948.